

BM2000 Control Systems

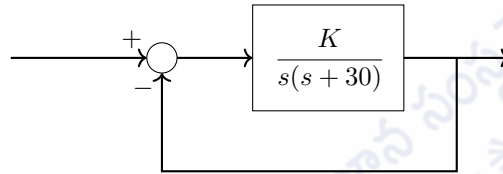
Problem set 2

Instructions

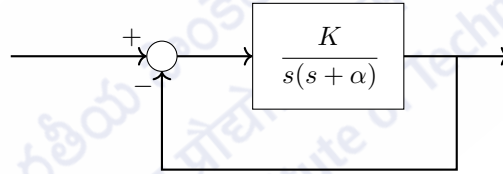
- You are not expected to submit the answers to these questions.

Questions

- For the system shown below, find the value of K that yields 10% overshoot for a step input.



- For the system shown below, find K and α to yield a settling time of 0.12 second and a 20% overshoot.



- For the following second order system

$$\ddot{y}(t) + 2\zeta\omega_n\dot{y}(t) + \omega_n^2 y(t) = \omega_n^2 x(t) \quad (1)$$

prove that the system is BIBO stable if $\zeta > 0$ and $\omega_n > 0$.

